

The FindDialog constructor is typical of Qt widget classes. The parent parameter specifies the parent widget. The default is a null pointer, meaning that the dialog has no parent.

The signals section declares two signals that the dialog emits when the user clicks the Find button. If the Search backward option is enabled, the dialog emits findPrevious(); otherwise, it emits findNext().

The signals keyword is actually a macro. The C++ preprocessor converts it into standard C++ before the compiler sees it. Qt::CaseSensitivity is an enum type that can take the values Qt::CaseSensitive and Qt::CaseInsensitive.

In the class's private section, we declare two slots. To implement the slots, we will need to access most of the dialog's child widgets, so we keep pointers to them as well. The slots keyword is, like signals, a macro that expands into a construct that the C++ compiler can digest.

For the private variables, we used forward declarations of their classes. This was possible because they are all pointers and we don't access them in the header file, so the compiler doesn't need the full class definitions. We could have included the relevant header files (<QCheckBox>, <QLabel>, etc.), but using forward declarations when it is possible makes compiling somewhat faster.

- finddiaalog.cpp

We will now look at finddialog.cpp, which contains the implementation of the FindDialog class and we also be providing the interpretations side-by-side, since the code is quite long

- 1 #include <QtGui>
- 2 #include "finddialog.h"
- 3 FindDialog::FindDialog(QWidget *parent)

First, we include <QtGui>, a header file that contains the definition of Qt's GUI classes. Qt consists of several modules, each of which lives in its own library. The most important modules are QtCore, QtGui, QtNetwork, QtOpenGL, QtScript, QSql, QtSvg, and QtXml. The <QtGui> header file contains the definition of all the classes that are part of the QtCore and QtGui modules. Including this header saves us the bother of including every class individually.

In finddialog.h, instead of including <QDialog> and using forward declarations for QCheckBox, QLabel, QLineEdit, and QPushButton, we